MATH SURVEY

STUYVESANT HIGH SCHOOL

SPRING 1987

Math Survey is the official publication of the Mathematics Department of Stuyvesant High School. It represents an attempt to communicate the beauty and joy of mathematics to the high school community.

Cover design by Ira Sher.

We dedicate this magazine to Mr. Phillips, who gave us his time, his support and a pizza, just when we needed it most.

Math Survey

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Special thanks to Mr. Clancy, Mr. Geller, Mr. Greenfield and Mr. Leonardo for their assistance.

Covering of nxn Boards
by Michael Hutchings

Let us start with an old problem: Is it possible to exactly cover an 8x8 chessboard from which two opposite corners have been removed (Figure 1) with dominoes?

Figure 1

The answer is no!

If we count, we find that there are 30 black squares and 32 white squares. Each domino must cover one white square and one black square. After placing 30 dominoes, we must have two white squares left. There will be no way to place a domino so as to cover both.

The next problem is slightly more difficult: Show how to tile with T-tetraminoes (Figure 2) all nxn boards that can be so tiled and prove that it is impossible to so tile any others.

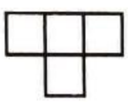


Figure 2

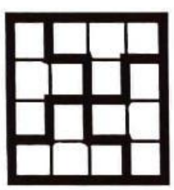


Figure 3

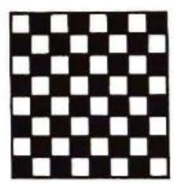


Figure 4

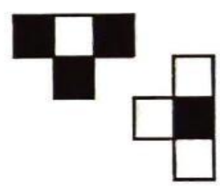


Figure 5

First, we note that we can tile the 4x4 square (Figure 3). Whenever n is a multiple of 4, we can divide the board into 4x4 squares, which we can tile. Thus, if n is a multiple of 4, the board can be tiled.

It would be nice to show that if n is not a multiple of 4, then the nxn board cannot be tiled. We know that our tetramino consists of four squares. If k tetraminoes exactly cover the nxn board, $n^2 = 4k$ is a multiple of 4 and n must be even.

We then color our board diagonally with two colors, checkerboard-style (Figure 4 illustrates the case n=8). We see that the tetramino covers three squares of one color and one of the other, regardless of orientation (Figure 5). On any nxn board where n is even, there are equal numbers of white and black squares. To cover every square, we must cover equal numbers of white and black squares. Whenever we place a tetramino on the board, we cover two more squares of one color than the other and need to put down another tetramino with the opposite coloring to even it out. Our final tiling must contain an even number of tetraminoes. Therefore n^2 must be a multiple of 8, and n must be a multiple of 4.

We have eliminated all n except multiples of 4, which we know can be tiled.

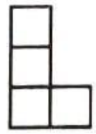
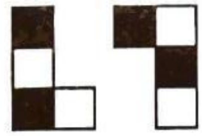
  

Figure 6 Figure 7 Figure 8

Another polyomino we can try is the L-tetramino (Figure 6). We can tile a 4x4 square, as in Figure 7. We can therefore tile any nxn board where n is a multiple of 4. If we try to to duplicate our previous proof and color the board with two colors the way we did before, we find to our frustration that each tile covers two white squares and two black squares (Figure 8). Let's try coloring the board with stripes instead (Figure 9 shows the case n=8). Now our tetramino covers three squares of one color and one square of the other, no matter how it is oriented (Figure 10). With this, the rest of the proof is the same as before.

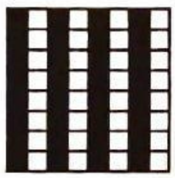
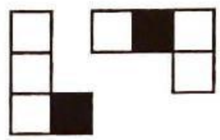
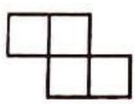
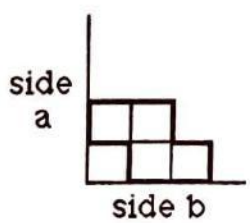
 

Figure 9 Figure 10

Yet another polyomino is the skew tetramino (Figure 11). If you try to cover any square with it, you will probably find the problem quite annoying. It happens to be impossible to do so.

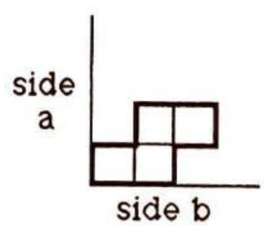


Figure 11 Figure 12 Figure 13

To show this, we start at one corner of the board. There is only one way to place a skew tetramino adjacent to side a and b that doesn't leave an isolated, unfillable square, as in Figure 12. The correct placement is shown in Figure 13. Then, the only way to fit in another skew so that we don't leave one or two unfillable squares is shown in Figure 14. We continue this way across.

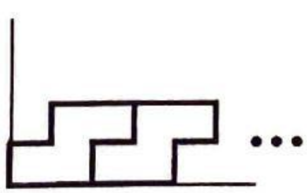
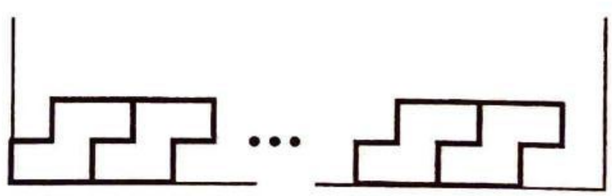
 

Figure 14 Figure 15

(continued on page 25)

A Quiet Conspiracy

by Alexander Ng

translated from the Greek by Cassidy Curtis

There is a quiet conspiracy at hand. For the past twenty years Stuyvesant High School has fostered a misconception that most students accept unquestioningly. What is this long running hoax? That a working knowledge of the elementary calculus, or "AP Calc" as it has been rechristened, is essential to most respectable professional and technical careers (business, medicine, architecture, et cetera).

The advent of the computer has made this less than truthful. There was a time, decades ago, when managers, scientists, and technicians had to sit down with pencil and paper, grinding out tedious derivatives and integrals of elementary functions. But that is long past, and the need for AP calculus has diminished accordingly.

Software companies are putting out programs that solve all sorts of complicated differential equations. Indeed, you can purchase cheap hand-held calculators able to both differentiate and integrate. Be prepared: soon enough, you can be sure some clever entrepreneur will concoct a software package entitled "AP Calc Homework in 15 Minutes or Your Money Back." Even before this automation existed, an abundance of handbooks had excised the thinking aspect of any calculus problem. All you really needed to do was look up the appropriate formula and do simple plug-in math.

You, the reader, might ask, "Why learn calculus? What's the use?" In a nutshell, an appreciation of the calculus, its significance, and its main assertions is necessary for any thorough liberal education. No one is exempt. Indeed, every reasonably good university requires calculus as a freshman mathematics course. Ever wonder why?

As a branch of math, the calculus is unique. It bridges the gap between pre-calculus mathematics, material old as the ancient Greeks, and the newer, more abstract advances of the eighteenth century and onward. High school students can tackle the calculus with reasonable ease.

Its history also traces out a weird and exemplary path. Two eccentric geniuses, Isaac Newton (the famous physicist and mathematician) and Gottfried Leibnitz (the German philosopher), were responsible for creating the calculus. Working concurrently but independently, they welded differential and integral calculus into a unified framework of thought and symbolic notation. Leibnitz invented his version between 1673 and 1676; Newton published a sketchy explanation of his results in 1687, years after he discovered them. A controversy raged over possible plagiarism. The English claimed Newton to be the true inventor, and Leibnitz a fraud, while the Europeans supported their German originator. In protest, the English repudiated Leibnitz's more convenient symbolic notation (i.e. the familiar $^{dy}/_{dx}$ and $\int f(x)dx$), favouring instead Newton's ugly fluxions and fluents. (A fluxion, or derivative of y is $\dot{y}$. Its fluent, or integral is $\overset{\square}{y}$.)

At the time of its creation the calculus was a flimsy substance indeed. Newton's logic contained obvious flaws. His critic, Bishop George Berkeley, sarcastically described fluxions as "ghosts of departed quantities." In the years that followed, Berkeley's indictment was borne out.

Bizarre and contradictory results cropped up everywhere as the calculus was recklessly misapplied.

Much of the problem rested on an inadequate definition of the limit (LIM f(x)). It took over a century of contemplation and the conceited French engineer Augustin-Louis Cauchy to put the calculus on sure footing. The infamous *delta-epsilon* definitions of limit, continuity, and definite integral were all his doing.

Within our own lifetimes the calculus has been thoroughly overhauled. During the 60's, mathematicians reinvented calculus without the complicated notion of the limit. Using something called hyperreals, nonstandard analysis provided an alternative to the traditional limits approach.

Aside from its colorful history, calculus was a genuine breakthrough. Before its creation, 5000 years of mathematics had only been able to describe static quantities. A grain merchant had only been able to sell wheat in whole bushels or discrete fractions of bushels. With the calculus, he could sell wheat straight from the grain silo by measuring the flow of grain pouring down the chute. In essence, calculus was a study of change and its application to tangible quantities. It permitted astronomers to calculate the trajectories of planets, engineers to measure the amount of force acting on a moving body, and mathematicians the exact area under any continuous curve. It made life a lot easier.

These are some aspects of calculus that should (and could) be discussed in elementary calculus classes. As it stands now, only mathematics majors, who take advanced courses, can apppreciate these concepts. It should matter little if students ever become mathematicians. While mathematical ideas become progressively advanced, their creative processes don't change. Students ought to be exposed to mathematical thinking just as they ought to learn freshman college English, chemistry, and calisthenics 101. When they are not, the humanistic and intellectual value of calculus can be lost. The subject becomes another means for grinding out a bachelor's degree. While a college education is definitely an eye-opening experience, it seems that too many of us will graduate with only one eye opened.

Taking calculus in high school is not without its merits though. Since professors curve the test scores, students who spend a year in AP Calc will inevitably perform better than those who haven't, which could spell the difference between an A and a C in the class.

Stuyvesant is one of several high schools across the nation offering a calculus course that teaches towards this higher purpose. Stuyvesant's M11C and M11D sequence (i.e. third term calculus and differential equations) does not attempt an encyclopedic coverage of topics, and there is no college-bound exam to cram for. More importantly, by presenting the ideas in a more informal discussion atmosphere, these courses expose students to the kind of environment in which mathematics is actually created. A problem is presented and students seek a successful solution or explanation. The instructor goads and hints to them a little, but students are not so much instructed as they are enlightened.

Newton's fluxions and fluents have outlasted his more famous laws of physics. Without a doubt, calculus is an intellectual triumph of the first order. Long after Sanford Ganzell forgets how to maximize the volume inside a cardboard box, he will appreciate and understand the richness of the calculus and its theoretical descendants.

Translations and Rotations
by Holly Engelman

Perhaps the most common method of graphing equations is to pick points and estimate. This technique is simple to understand and easy to use, at least on uncomplicated equations. However, if our equation is:

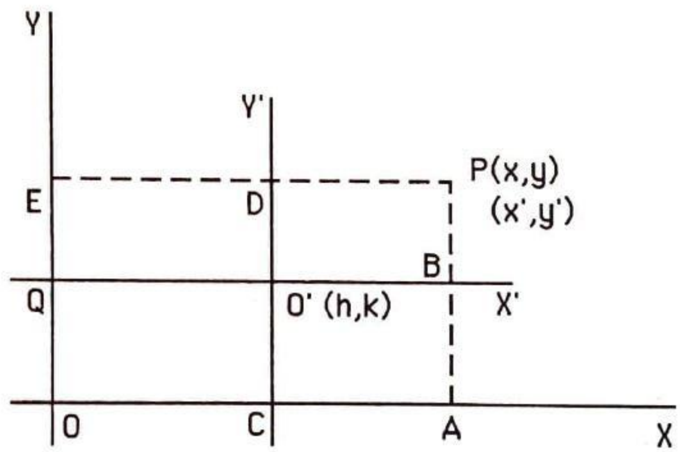
$$41x^2 - 24xy + 34y^2 = 25,$$

we may not feel very secure about the "pick-points-and-estimate" method.

However, in this modern day and age, there are ways of getting around this problem. Nearly all of these complex second-degree equations can be simplified into the tractable forms that we have been taught, whether they be hyperbolæ, ellipses, or parabolæ. When the equation is graphed we shall find that almost all of the properties of the curve have been preserved; only its location on the plane will have been altered.

Some complex equations are the result of a motion known as a translation. This simply shifts the axes a known distance to the side and up or down from their original position. The other motion commonly seen is called a rotation. It rotates the axes a certain number of degrees, causing a difference in the position of the curve when it is looked at with regard to the new axes.

In order to transform unpleasant equations into ones we understand, we must first derive formulæ relating the given equation to the new equation we will assemble.



To derive the formulæ for translation, we draw the original axes X and Y and another set of axes parallel to the first, called X' and Y'. We then take a point P which has two names under the different axes, (x,y) and (x',y'). From this figure we can determine that:

$$x = EP, \qquad x' = DP, \qquad h = OC = ED,$$
$$y = AP, \qquad y' = BP, \qquad k = CO' = AB.$$

We see that:

$$EP = ED + DP \quad \text{and}$$
$$AP = AB + BP.$$

Therefore,

$$x = x' + h \quad \text{where h equals the horizontal change, and}$$
$$y = y' + k \quad \text{where k equals the vertical change.}$$

What can we do with these formulæ? Well, now that we know x and y in terms of x' and y', we can substitute these values back into our original equations. Remember that h and k are the distances the axes have been moved horizontally and vertically. Also note that if the curve is symmetric with respect to the new origin, there can be no first degree terms. However, there is a first degree term in the equation for the parabola. In simplifying the equation we substitute the formulæ and set coefficients of x' and y' to zero. Watch...

$$\begin{aligned} y^2 + 8x + 4y - 20 &= 0 \\ (y' + k)^2 + 8(x' + h) + 4(y' + k) - 20 &= 0 \\ y'^2 + 2y'k + k^2 + 8x' + 8h + 4y' + 4k - 20 &= 0 \\ y'^2 + k^2 + (2k + 4)y' + 8x' + 8h + 4k - 20 &= 0 \\ 2k + 4 &= 0 \\ k &= -2 \\ k^2 + 8h + 4k - 20 &= 0 \\ 4 + 8h + (-28) &= 0 \\ 8h &= 24 \\ h &= 3 \end{aligned}$$

We have just found that the curve contains a first degree x term. Hence, it must be a parabola. We have also found that the axes must be translated (+3,-2), three to the right and two down. Now we must find the equation to graph. Again we substitute the formulæ into the equation, this time knowing h and k.

$$\begin{aligned} y^2 + 8x + 4y - 20 &= 0 \\ (y' - 2)^2 + 8(x' + 3) + 4(y' - 2) - 20 &= 0 \\ y'^2 - 4y' + 4 + 8x' + 24 + 4y' - 8 - 20 &= 0 \\ y'^2 &= -8x' \end{aligned}$$

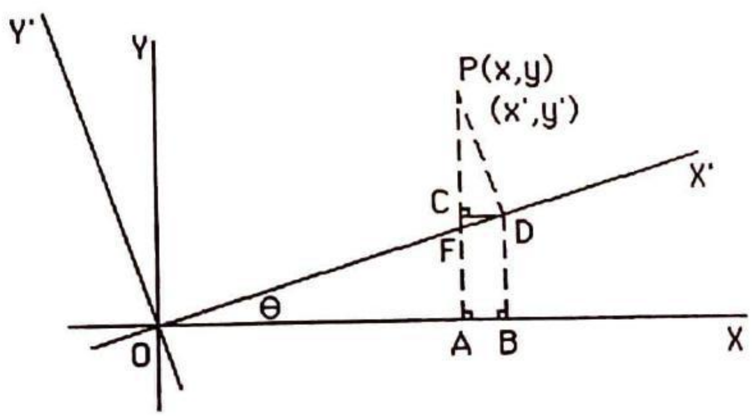
A parabola equation! How sweet and simple this looks! Graphing is up to you.

Now, let's try the equation: $x^2 + y^2 + 4xy = 16$.

$$\begin{aligned} x^2 + y^2 + 4xy &= 16 \\ (x' + h)^2 + (y' + k)^2 + 4(x' + h)(y' + k) &= 16 \\ (x' + h)^2 + (y' + k)^2 + \mathbf{4x'y'} + 4hy' + 4kx' + hk &= 16 \end{aligned}$$

We know of no "regular" formula using x'y', yet in this equation, x'y' can not be discarded. From this we conclude that when we have an equation with xy, the translation formulæ don't work. Therefore, we must derive another set of formulæ, the rotation formulæ.

We begin by drawing old axes X and Y and new axes, X' and Y', all stemming from O, the origin. θ is the angle of rotation of the axes.



We see that:

$$\begin{aligned} x &= OA, & x' &= OD, \\ y &= AP, & y' &= DP. \end{aligned}$$

We note that angle CPD is equal to θ, from similar right triangles CPD and AOF. Therefore we can conclude that:

$$CD = y' \sin\theta, \qquad OB = x' \cos\theta,$$
$$CP = y' \cos\theta, \qquad BD = x' \sin\theta.$$

Since we know that:

$$x = OB - CD,$$
$$y = BD + CP,$$

we can then substitute to obtain:

$$x = x'\cos\theta - y'\sin\theta$$
$$y = x'\sin\theta + y'\cos\theta$$

These are the rotation formulæ! However, to apply them we must learn one more thing.

Because we want to eliminate the xy term when we use rotation, we must find the correct angle for θ which will eliminate that term. When we substitute the formulæ into the general equation for a conic,

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0,$$

the coefficient of the x'y' term turns out to be :

$$-2A\sin\theta\cos\theta + B(\cos^2\theta - \sin^2\theta) + 2C\sin\theta\cos\theta$$

which equals:

$$-A\sin 2\theta + B\cos 2\theta + C\sin 2\theta.$$

We set this equal to zero:

$$\begin{aligned} -A\sin 2\theta + B\cos 2\theta + C\sin 2\theta &= 0 \\ (C - A)\sin 2\theta + B\cos 2\theta &= 0 \\ (C - A)\sin 2\theta &= -B\cos 2\theta \\ -\frac{(C-A)}{B}\frac{\sin 2\theta}{\cos 2\theta} &= 1 \\ \frac{\sin 2\theta}{\cos 2\theta} &= \frac{B}{A - C} \\ \tan 2\theta &= \frac{B}{A - C} \end{aligned}$$

Using this formula with the half angle formulæ, we can find θ. The two rotation formulae above can now be applied once the angle is found.

So to recap: Given any second-degree equation, no matter how messy, we can first rotate and then translate so that it becomes a nice and simple hyperbola, parabola, or ellipse.

Regents Refresher

Due to the New York State Board of Regents' introduction of a new high school mathematics curriculum, it is more difficult for us to provide overview of the material covered than in previous years. Readers who are in the classic courses—algebra, geometry, and trigonometry, designated M1/2, M3/4, and M51/52—should have no difficulty in recognising the material required for their exams. For the benefit those in sequential math—MQ1/2, MQ3/4, and MQ5/6— we have prefixed the appropriate sections with roman numerals, thus indicating the year in which the material is taught.

This section does not pretend to be a complete review. It includes only the most important theorems and formulæ.

(II,III) <u>Equations of Plane Curves</u>:
Line: $y=mx+b$ where $m=$ slope and $b=$ y-intercept.
$y-y_1= m(x-x_1)$ where (x_1,y_1) is a point on the line and m is the slope.
Slopes of parallel lines are equal.
The product of the slopes of two perpendicular lines is -1.
Circle: $(x-h)^2 + (y-k)^2 = r^2$, where (h,k) is the center of the circle with radius r.
Ellipse: $x^2/a^2 + y^2/b^2 = 1$, where a and b are the major and minor axes.
Parabola: $(x-h)^2 = 2p(y-k)$, where (h,k) is the vertex and p is the distance from the vertex to the directrix
or $y = ax^2 + bx +c$, where $x = -b/2a$ is the axis of symmetry.
Hyperbola: $x^2/a^2 - y^2/b^2 = 1$, where a and b are the major and minor axes.

(III) <u>Logarithms:</u> "$\log_b N=E$" <u>means</u> $b^E=N$.

$\log_c 1=0$ $\quad$ $\log_c c=1$ $\quad$ $\log_c c^b=b$

$\log_c a^b=b \log_c a$ $\quad$ $\log_c ab=\log_c a + \log_c b$ $\quad$ $\log_c 1/a=-\log_c a$

$\log_c(a/b)=\log_c a-\log_c b$

$\log_c \sqrt[b]{a}=(\log_c a)/b$

$\log_b a=\log_c a/\log_c b=\log_b c/\log_a c$

(II) <u>Laws of Exponents</u>:

$x^m \cdot x^n = x^{m+n}$ $\quad$ $(x^m)^n=x^{mn}$ $\quad$ $x^m/x^n=x^{m-n}$ $\quad$ $x^{-m}=1/x^m$ $\quad$ $x^0= 1$

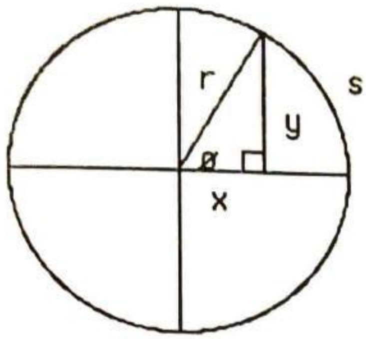
$x^{1/m}=\sqrt[m]{x}$ $\quad$ $x^{m/n}=\sqrt[n]{x^m}=(\sqrt[n]{x})^m$

(II) <u>The Quadratic Formula:</u>
The roots of the equation $ax^2 + bx + c$ are

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

If the discriminant, $b^2 - 4ac$, is positive: the roots are real and distinct.
zero: the roots are real and equal.
negative: the roots are complex conjugates.

(III) Trigonometry:



$s = r\emptyset$

$r^2 = x^2 + y^2$

SOH-CAH-TOA

$\sin\emptyset = y/r$
$\cos\emptyset = x/r$
$\tan\emptyset = y/x$
$\csc\emptyset = r/y = 1/\sin\emptyset$
$\sec\emptyset = r/x = 1/\cos\emptyset$
$\cot\emptyset = x/y = 1/\cot\emptyset$

$\sin^2\emptyset + \cos^2\emptyset = 1$
$\tan^2\emptyset + 1 = \sec^2\emptyset$
$\cot^2\emptyset + 1 = \csc^2\emptyset$
$\sin\emptyset = \cos(90°-\emptyset)$
$\tan\emptyset = \cot(90°-\emptyset)$
$\sec\emptyset = \csc(90°-\emptyset)$

Sum/Difference Formulæ:
$$\sin(x\pm y) = \sin x\cos y \pm \sin y\cos x$$
$$\cos(x\pm y) = \cos x\cos y \mp \sin x\sin y$$
$$\tan(x\pm y) = \frac{\tan x \pm \tan y}{1\mp\tan x\tan y}$$

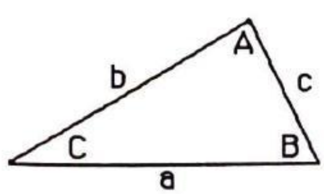
Double Angle Formulæ:
$$\sin(2x) = 2\sin x\cos x$$
$$\cos(2x) = \cos^2 x - \sin^2 x = 2\cos^2 x - 1 = 1 - 2\sin^2 x$$
$$\tan(2x) = \frac{2\tan x}{1 - \tan^2 x}$$

(Note: the double angle formulæ can be derived from the sum formulæ by letting x=y.)

Half Angle Formulæ:
$$\sin^2(x/2) = \frac{1-\cos x}{2} \qquad \cos^2(x/2) = \frac{1+\cos x}{2}$$
$$\tan^2(x/2) = \frac{1-\cos x}{1+\cos x}$$

In the graph of $y = N\sin(rx)$, the amplitude is N, and the period is $2\pi/r$.
360 degrees = 2π radians.
To convert from degrees to radians, multiply by $\pi/180$.
To convert from radians to degrees, multiply by $180/\pi$.

Triangles:



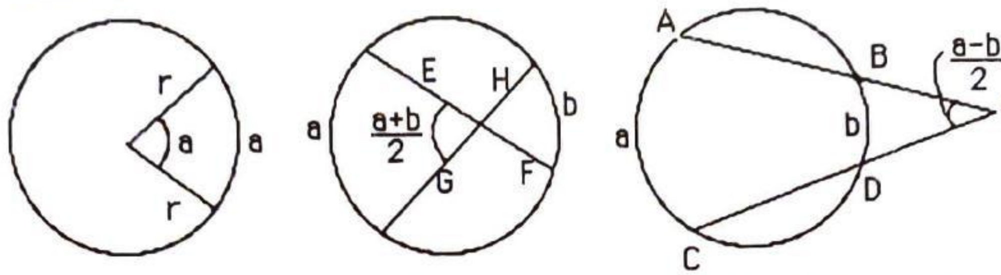
$$\text{Area} = (1/2)ab\sin C = (1/4)\sqrt{(a+b+c)(a+b-c)(b+c-a)(c+a-b)}$$

Law of Sines: $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} =$ diameter of circumcircle

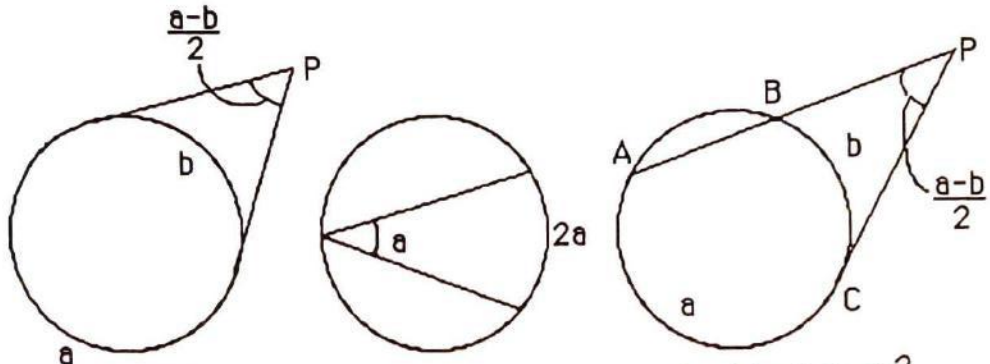
Law of Cosines: $c^2 = a^2 + b^2 - 2ab\cos C$

Law of Tangents: $\frac{\tan((A+B)/2)}{\tan((A-B)/2)} = \frac{a+b}{a-b}$

(III) Geometry:



r=radius EF=GH (PA)(PB)=(PC)(PD)



PA=PB (PA)(PB)=$(PC)^2$

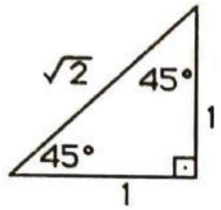
The required proofs for the Geometry Regents may be found in the Dressler text, pp. 733-737. The following are the numbers of the required proofs; 3,5,6,7,8,9,10,11,12,13.

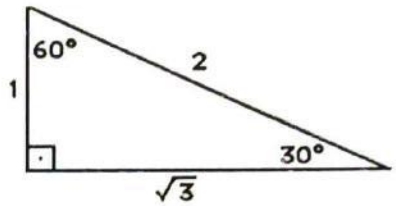
(II) (A=Angle, S=Side, b=base, h=altitude)

Two triangles are congruent if you can prove SSS, SAA, SAS, or ASA.
Two triangles are similar if you can prove AA.
Two right triangles are congruent if you can prove their hypotenuses are congruent and one pair of legs are congruent.(HL)
The sum of the angles in a triangle is 180°.
The base angles of an isosceles triangle are congruent.
Area of triangle=$(^1/_2)bh$.
Area of a trapezoid= $(^1/_2)(b_1+b_2)h$.
Area of a parallelogram=bh.
Area of a regular n-gon=$(^1/_2)ap$ (where a=apothem, p=perimeter).
In a right triangle with legs a and b and hypotenuse c, $a^2+b^2=c^2$.
In a circle, the diameter that is perpendicular to a chord bisects the chord.
The locus of points at distance r from point P is the circle with center P and radius r.
The locus of points equidistant from two points P and Q is the perpendicular bisector of PQ.
The locus of points equidistant from two lines l and m are the bisectors of the angles formed by l and m.

(B=area of base, h=height, r=radius)
Volume of a prism = Bh
Volume of a cylinder = πr^2h
Volume of a pyramid = $(^1/_3)$ Bh
Volume of a cone = $(^1/_3)\pi r^2h$
Volume of a sphere = $(^4/_3)\pi r^3$

The required constructions for the Geometry Regents may be found in the Dressler text, pp. 616-639. The following are the numbers of the required constructions; 5-7, 12, 13, 18-22, 25, 27-29.





(I, III) Probability:

Let A and B represent events in a sample space.

If A and B are independent, then $P(A \wedge B) = P(A) \cdot P(B)$.

If A and B are disjoint, then $P(A \wedge B) = 0$ and $P(A \vee B) = P(A) + P(B)$.

If the probabilities are uniformly distributed, event A can occur in $n(A)$ ways, and there are $n(S)$ possible outcomes in S, then $P(A) = n(A)/n(S)$.

The number of permutations of n objects is $n! = n(n-1)...(2)(1)$.

The number of permutaions of n objects, k of which are indistinguishable, is $n!/k!$.

The number of permutations of r objects chosen from a set of n objects is

$$_nP_r = n!/(n-r)! = n(n-1)(n-2)\cdots(n-r+1).$$

The number of combinations of r objects chosen from a set of n objects is

$$_nC_r = n!/r!(n-r)! = {_nP_r}/r!.$$

(III) The binomial theorem:

$$(a+b)^n = \sum_{r=0}^{n} {_nC_r}\, a^r b^{(n-r)} = a^n + {_nC_1}a^{n-1}b + {_nC_2}a^{n-2}b^2 + \cdots + {_nC_{n-1}}a^1b^{n-1} + b^n$$

(I,II) Logic:

Laws of Inference	Symbolic Forms
1. Law of Detachment (Modus Ponens)	$[(p \to q) \wedge p] \to q$
2. Law of the Contrapositive	$(p \to q) \leftrightarrow (\sim q \to \sim p)$
3. Law of Modus Tollens	$[(p \to q) \wedge \sim q] \to \sim p$
4. Chain Rule (Law of Syllogism)	$[(p \to q) \wedge (q \to r)] \to (p \to r)$
5. Law of Disjunctive Inference	$[(p \vee q) \wedge \sim p] \to q$ $[(p \vee q) \wedge \sim q] \to p$
6. Law of Double Negation	$\sim(\sim p) \leftrightarrow p$
7. De Morgan's Laws	$\sim(p \wedge q) \leftrightarrow (\sim p \vee \sim q)$ $\sim(p \vee q) \leftrightarrow (\sim p \wedge \sim q)$

8. Law of Simplification $(p \wedge q) \rightarrow p$

$(p \wedge q) \rightarrow q$

9. Law of Conjunction $(p) \wedge (q) \rightarrow (p \wedge q)$

10. Law of Disjunctive Addition $p \rightarrow (p \vee q)$

(I, III) Statistics

Measures of Central Tendency:

The mean, or arithmetic mean, of a set of n numbers is the sum of the numbers divided by n. Therefore, for a sample of scores, $x_1, x_2, x_3, \ldots, x_n$:

$$\text{The mean} = \bar{x} = \frac{x_1 + x_2 + x_3 + \ldots + x_n}{n}$$

When a set of data is arranged in numerical order, the "middle" score is called the median. If there is an even number of terms, then the median is found by averaging the two "middle" terms.

The mode is the score that appears most often in a set of data. For some sets of data there can be more than one mode (when two modes appear, the data is called bimodal) and, in some cases, no mode whatsoever.

Measures of Dispersion:

The range of a set of data is the difference between the highest and the lowest scores.

The mean absolute deviation of a set of data is the average value of the difference between the data and the mean. In symbols:

$$\text{mean deviation} = (1/n) \sum_{i=1}^{n} |x_i - \bar{x}|$$

The variance, v, of a set of data is the average of the squares of the deviations from the mean. In symbols:

$$\text{variance} = v = (1/n) \sum_{i=1}^{n} (x_i - \bar{x})^2$$

The standard deviation, s, is a set of data is equal to the square root of the variance. In symbols:

$$\text{standard deviation} = s = \sqrt{v}$$

Lifestyles of the Rich and Mathematically Inclined

by Pheth Goch

We're coming to you here from fabulous Stuyvesant High School, home of the world-famous Stuyvesant Math Team. We'll be talking to members of the team and finding out just what they do to relax after a hard day's doing... whatever it is they do.

The principle home-away-from-home of the math team is the posh environment of Room 517. This room was personally decorated at astronomical cost by Coach Geller. The chairs are covered with genuine Nordic leather and the desks are real mahogany. Coach Geller believes that the mathematical mind works best in a comfort-filled environment. "We tried to get a hot tub installed for the team, but too many teachers in those other departments thought they couldn't teach with the steam it would create." Better luck next time, Coach Geller.

Similar measures are taken during meets. The team arrives in big, beautiful stretch limousines provided by the department. Upon arrival, each member is ushered into their own customized meet room, with (almost) every luxury they could ask for. There have been problems trying to get a wave pool installed for the senior team captain, Zeph Landau. "We think Zeph's scores would shoot way up if we could just arrange it so that he could do his math problems while on his surfboard," commented Coach Geller.

But boy, oh boy, is Elizabeth Wilmer, the other senior captain, in heaven. "There are 24,310 different colored bottles of nail polish in my room, and I go through most of them during a meet." Ms. Wilmer brandished her fingernails at me and I was almost blinded by the variety and brightness of of the colors. Absolutely incredible!

Outfitting future captain Sanford Ganzell's room turned out to be the most challenging. "Sandy has always been a very energetic person, especially when he's thinking hard, and it shows, physically.", said math department chairperson and arch-fan, Dr. Richard Rothenberg. "We were worried that he might hurt himself during the tough meets, so for his own sake we put rubber padding on the walls of his room." Team members are still at a loss to explain Sandy's demand for mirrors on the ceiling.

After the meets, the team goes out to celebrate at such glamorous places as Lutece and the Palladium, all paid for with the proceeds from bake sales. "It's really an incredible scene," commented senior and team member Jen-Ling Liu. "Can you just imagine it when they unleash the entire team on these places!" The damage charges incurred by the math team in their victory parties during this last season is estimated to be almost equal to the national debt.

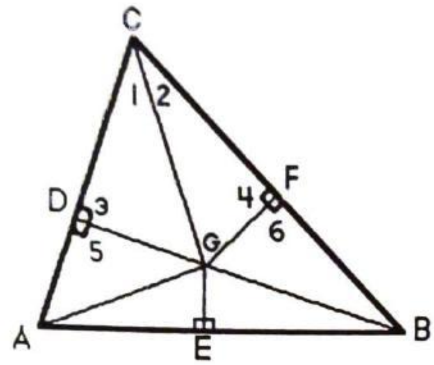
When asked if the perks and luxuries lavished upon the math team in any way detracted from the rest of the school, Principal Abraham Baumel replied: "I really don't think so, and anyway, this gives the other students role models to look up to for guidance. Besides, where our teams are concerned, money is no object; after all, look at the debate team."

Yes, it's a good life on the Stuyvesant math team. There's nothing that can't be done for that person with the incredibly high ARML scores. "We've been thinking about a few statues of famous math team captains past in the new building." says Mr. Baumel. Prepare yourselves math fans, for the immortalization of Darien Lefkowitz.

This is Pheth Goch, signing off, from incredible Stuyvesant High School, with caviar wishes and champagne dreams.

Find the Flaw

The mathematical proof is the oldest form of logical argument and deductive reasoning. This is most clearly evident in the field of geometry where given hypotheses are used to find a conclusion. But are the conclusions always "certain"? Sometimes, the absurdly impossible is proven true by flawed reasoning. Below, the reader will see the conclusion that all triangles are isosceles, that is, all triangles have two congruent sides. The proof must have a mistake. Can you find the flaw?



<u>Given:</u>
Any ΔABC

<u>Prove:</u>
ΔABC is isosceles

(1) Construct bisector of $\angle C$

(2) Construct $\perp$ bisector of AB

(3) They meet at point G.

(4) From G drop a perpendicular to AC and BC, namely GD and GF

(5) hence $\angle 1 \cong \angle 2$
$\angle 3 \cong \angle 4$
$CG \cong CG$

(6) $\Delta CDG \cong \Delta CFG$ (AAS $\cong$ AAS)

(7) then $DG \cong GF$
$\angle 5 \cong \angle 6$
$AG \cong GB$ (Point G lies on $\perp$ bisector of AB: $\therefore$ G is equidistant from endpoints A and B)
$\Delta ADG \cong \Delta BFG$ (hyp leg $\cong$ hyp leg)

(8) hence $CD \cong CF$
$DA \cong FB$

(9) $CD + DA \cong CF + FB$

(10) $CA \cong CB$

(11) $\therefore$ ΔABC is isosceles

Investigation of Two Prime Number Conjectures Through Calculus

by N. Clemente

It is an interesting fact that

$$\pi(x + y) \leq \pi(x) + \pi(y) \qquad \ldots (1)$$

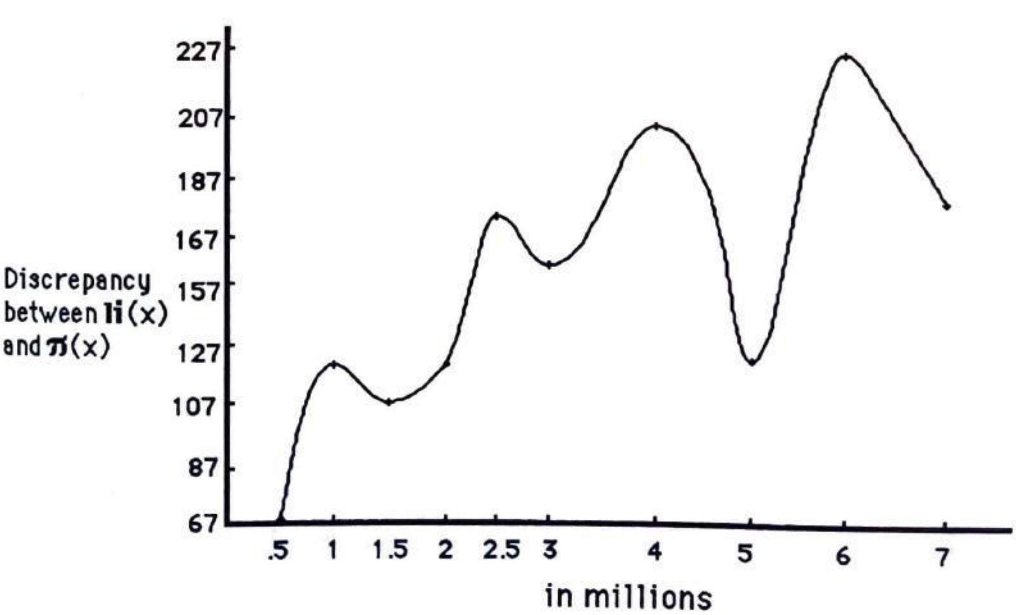
where $\pi(x)$ represents the number of primes not exceding an integer x, has been neither proven nor disproven and is therefore a conjecture. At first glance its validity seemed obvious; since the frequency of prime numbers decreases, it appeared that the number of primes under the sum of two integers should be less than (or *maybe* equal to) the sum of the number of primes under the individual integers.

To examine the conjecture we need a numerical way to express $\pi(x)$. A remarkable expression called the logarithmic integral exists, which yields the approximate number of primes not exceeding a given limit and is given by

$$\mathbf{li}(x) = \int_2^x \frac{dt}{\log(t)} \qquad \ldots \qquad (2)$$

At relatively small values, the approximation is very inaccurate. But for larger values, the $\mathbf{li}(x)$ tends to $\pi(x)$. This is very convenient, as the validity of the conjecture is questionable at these values. So by expressing the conjecture in terms of the approximation, we can perhaps determine whether it will hold true for large values. (see figure 1).

Figure 1



We now rewrite the conjecture in terms of the approximation:

$$\int_2^{x+y} \frac{dt}{\log(t)} \leq \int_2^{x} \frac{dt}{\log(t)} + \int_2^{y} \frac{dt}{\log(t)} \qquad \ldots (3)$$

where x and y are integers. Let us choose values of x and y such that $y > x$. Since the function is continuous for $t>0$ we may split the integral on the left of the inequality:

$$\int_2^{x+y} \frac{dt}{\log(t)} = \int_2^{x} \frac{dt}{\log(t)} + \int_x^{y} \frac{dt}{\log(t)}$$

Substituting, we obtain

$$\int_2^{x} \frac{dt}{\log(t)} + \int_x^{x+y} \frac{dt}{\log(t)} \leq \int_2^{x} \frac{dt}{\log(t)} + \int_2^{y} \frac{dt}{\log(t)}$$

Cancelling, we finally obtain

$$\int_x^{x+y} \frac{dt}{\log(t)} \leq \int_2^{y} \frac{dt}{\log(t)} \qquad \ldots (4)$$

or, asymptotically,

$$\int_x^{x+y} \frac{dt}{\log(t)} \leq \pi(y) \qquad \ldots (5)$$

both of which are true *whenever* (1) *is true*.

We need only to examine the conjecture for $y > x$, since the values thus obtained will also yield values for $x > y$. This is true because x and y are arbitrary. If we were to pick two integers, one being greater than the other, label the smaller one x and the larger y, and find a value for the integral, that value must be the same as that found when we call the larger one x and the smaller y, since the numbers themselves have not changed. To examine the integrals for $x > y$ would therefore be a waste of time and effort. We must, however, examine values of $x = y$.

A comparison of the integral in (5) and **li**(y) shows that the integral is indeed less than **li**(y) on the given interval. Due to errors in the approximations, however, the information found is relatively inaccurate. A trend can nevertheless be seen that the difference between the two increases with larger values. We can not be sure of the status at extreme values, but perhaps by examining a second conjecture we may draw further information.

Let us examine

$$\pi(x+y) < \pi(x) + 2\pi(y/2) \qquad \ldots (6)$$

We may use the same method to express (6) in terms of an integral approximation.

First we express the second conjecture as we did in (3), obtaining

$$\int_2^{x+y} \frac{dt}{\log(t)} \leq \int_2^{x} \frac{dt}{\log(t)} + 2\int_2^{y/2} \frac{dt}{\log(t)}$$

We now split the integral:

$$\int_2^x \frac{dt}{\log(t)} + \int_x^{x+y} \frac{dt}{\log(t)} \leq \int_2^x \frac{dt}{\log(t)} + 2\int_2^{y/2} \frac{dt}{\log(t)}$$

Cancelling, we obtain

$$\int_x^{x+y} \frac{dt}{\log(t)} \leq 2\int_2^{y/2} \frac{dt}{\log(t)}$$

or, asymptotically,

$$\int_x^{x+y} \frac{dt}{\log(t)} \leq 2\pi(y/2),$$

both of which are true *whenever* (6) is true. Note that the right side is identical to that found in (5).

A comparison between this integral and the difference between it and 2**li**(y/2) shows that the integral is indeed less. Further investigation shows, moreover, that the difference between the integral and 2**li**(y/2) is greater than the difference between the integral and **li**(y).

The difference between each succeeding y-value (for the same x-value) increases for both conjectures. The rate at which the difference increases is greater, however, for the second conjecture. The difference is also initially higher. At extremely high values, for which there are currently no tables, this difference may decrease, and the integral may approach **li**(y) and 2**li**(y/2). The values of 2**li**(y/2) are greater than those of **li**(y) for these values, however, since prime numbers become more scarce. Thus, if one of the conjectures were to be true, (6) *would be more likely to be true* than (1).

Again, the data is inaccurate. Perhaps with more accurate data, a more definite solution could be drawn. The accuracy of the data could be improved through several different methods. The approximations of $\pi(x)$ and other integrals could be improved through the use of Newton-Cotes formulæ other than Simpson's Rule (the one used to gather information for this article), by adding an accurate error function, by varying the limits of integration, or by expressing the integrand as an integral and perhaps varying its limits.

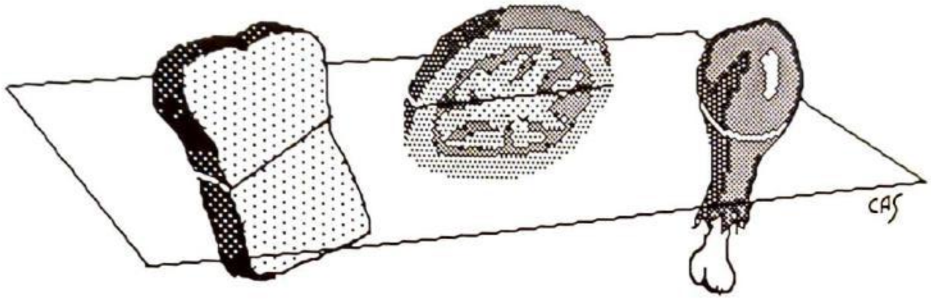
If you are interested in continuing research on the subject of prime numbers, you might want to read the article written by Darien Lefkowitz in the previous issue of *Math Survey*. Also, if your mathematical mettle is sufficient and your will strong, you may wish to read "Application of the Operational or Symbolic Calculus to the Theory of Prime Numbers" by B. Van der Pol in the Series 7, Volume 26 (December 1938) edition of *Philosophical Magazine,* pages 921 to 940. The article involves the Laplace Transformation and the Riemann-Zeta function, among other complex concepts.

On the Bisecting of a Ham Sandwich
by Sandy Ganzell

Theorem: The volumes of any three solids, A, B, C, of any size or shape, placed anywhere in space, can always be exactly and simultaneously bisected by a plane.

If we interpret A and B as slices of bread and C as a slice of ham between them, then the theorem can be restated as

Theorem: With one stroke of a knife, a ham sandwich can be divided so that both slices of bread and the ham are cut exactly in half.



Proof: We select a sphere S which encloses A, B and C. We know such a sphere exists because A, B and C are bounded. Let z be the center of S and r its radius. For each point $x \in S$, let L_x denote the diameter of S though x.

Lemma 1: For every $x \in S$, there is a unique point x_A on L_x such that the plane perpendicular to L_x at x_A divides A in half by volume.

Proof: For each point $y \in L_x$, let P_y denote the plane through y perpendicular to L_x, and let h(y) denote the volume of the part of A on the same side of P_y as x. As y varies from x' to x, h(y) varies from the volume of A down to zero. If y_1 and y_2 are in L_x, the difference $|h(y_1) - h(y_2)|$ is at most the volume of the part of the solid sphere between the planes Py_1 and Py_2, and this is less than $\pi r^2 |y_1 - y_2|$. This shows that h is continuous at each point y of L_x, by the definition of continuity. Thus, there exists a point y such that h(y) is half the volume of A. To prove the uniqueness of this point, assume that two such points existed. Then there would be two parallel planes P_1 and P_2 both dividing A in half. This leads to a contradiction and thus completes the proof.

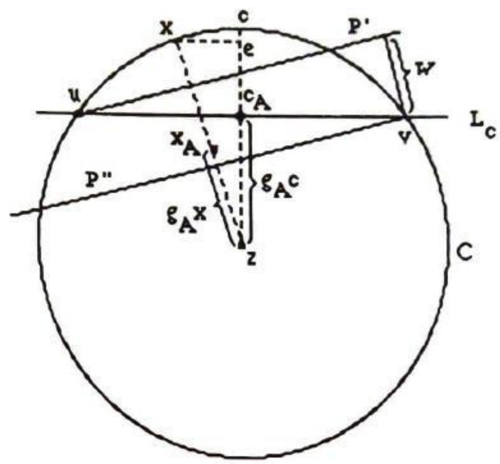
We now let $g_A x$ be the directed distance $d(z,x_A)$ with a positive sign if x_A is on the segment from z to x and with a negative sign if x_A is on the segment from z to the antipode x' of x. (If $x \in S$, the antipode x' of x is the other point in which the line through x and z meets S.) Since L_x and $L_{x'}$ coincide and have oppositely oriented coordinates, and since $x'_A = x_A$, it follows that $g_A x' = -g_A x$.

We define x_B, $g_B x$ and x_C, $g_C x$ symmetrically using B and C in place of A. Now consider the mapping $f: S \rightarrow R^2$ which assigns to each $x \in S$ the point with the coordinates:

$$f(x) = (g_A x - g_B x, g_A x - g_C x).$$

Lemma 2: f is continuous.

Proof: It suffices to prove the continuity of each coordinate of f. We shall prove only that g_A is continuous, and leave the rest to the reader. Let c be a point of S at which the continuity of g_A is to be proved, and let c_A be the point on L_c where the plane P_c perpendicular to L_c cuts A exactly in half. Let x be a point on S near c, and let x_A, P_x be defined similarly. We wish to show that $|g_A x - g_A c|$ can be made small (less than any challange number $\varepsilon>0$) by requiring x to be near c .



Let u and v be the points where the great circle contained in the plane though c, x, and z meets P_c. Let P' and P" be the planes perpendicular to L_x passing through u and v respectively. Let N denote the interior of S. The part of N on the same side of P' as c is included in the part of N on the same side of P as c. Since A is a subset of N, the part of A on the same side of P' as c is included in the part of A on the same side of P_c as c. So if V denotes the volume of A, the volume of the part of A on the same side of P' as c is at most $^V/_2$. By a similar argument, the volume of the part of A on the side of P" opposite to c is at most $^V/_2$. It follows that P_x must lie between P' and P". Therefore, if w is the distance between P' and P", we have

$$|g_A x - g_A c| < w.$$

An estimate of the size of w is obtained by noting from the similarity of two triangles that

$$\frac{w}{d(u, v)} = \frac{d(e, x)}{d(z, x)},$$

where e is the perpendicular projection of x on L_c. Since $r = d(z,x)$, this gives

$$w = \frac{d(u, v)}{r} d(e, x).$$

Since $d(u, v) \leq 2r$, and $d(e, x) \leq d(c, x)$, we obtain:

$$w \leq 2d(c, x).$$

Therefore:

$$|g_A x - g_A c| < 2d(c, x).$$

For a given $\varepsilon > 0$, take $\delta = \varepsilon/2$; then $x \in N(c, \delta)$ implies that $|g_A x - g_A c| < \varepsilon$. This shows that g_A is continuous at c. Since this holds for each $c \in S$, it follows that g_A is continuous on S. This completes the proof of our lemma.

We can now complete the proof of the theorem. K. Borsuk and S. Ulam proved in 1933 that every continuous mapping f: $S \to P$ of a sphere into a plane maps some pair of antipodes of S into the same point. Thus, for at least one pair q, q' of antipodes, $f(q) = f(q')$. Equating the coordinates of $f(q)$ and $f(q')$ yields

$$g_A q - g_B q = g_A q' - g_B q',$$
$$g_A q - g_C q = g_A q' - g_C q'.$$

Since $g_A q' = -g_A q$, $g_B q' = -g_B q$, $g_C q' = -g_C q$,

the first equation reduces to $g_A q = g_B q$

and the second to $g_A q = g_C q$.

Hence, for the point $q \in S$ whose image coincides with that of its antipode, we have $q_A = q_B = q_C$, and the plane perpendicular to L_x through this point divides all three regions in half. We have completed our proof.

(continued from page 6)

Since we're trying to tile an nxn square with tetraminoes, n is a multiple of two and we eventually end up in a position like Figure 15. The only way that we can fit another tetramino causes it to go over the edge, as in Figure 16.

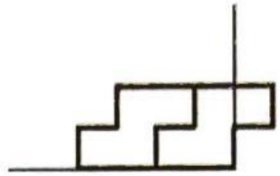


Figure 16

Clearly, this doesn't work. Oh, well.

For more of a challenge, you might delve into the realm of pentominoes, all 12 of which can fit neatly into the 5x12 rectangle shown in Figure 17.

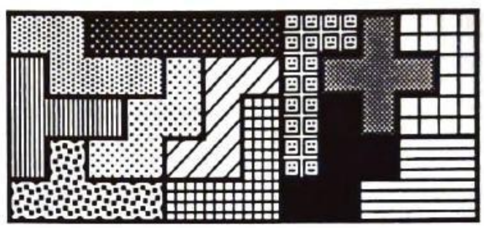


Figure 17

A Note on the Text

We are indebted to the Apple Computer Corporation for their generous donation of the Macintosh computers on which this magazine was produced. This magazine was printed with an Apple typeset quality LaserWriter printer on the premises of Stuyvesant High School. The text was set in the Times typeface; "Times" is a trademark, valid under applicable law, of the Allied Corporation and is used by Apple under license from Mergenthaler Linotype. The use of this trademark by the Mathematics Department of Stuyvesant High School does not entail any right, title, or interest in or to this trademark. Apple, Macintosh, and LaserWriter are trademarks of Apple Computer, Inc.

BOOSTERS

Solowey is vicious !!!
Earwaxtoecheeserunnynose!
This booster is dedicated
to the honorable Teru
Kuwayama. Come home soon!
DECORATE ME
Booster this be.
MRS ABRAMSON IS A SAINT.
SHE IS A GODDESS
LISA-I LOVE YOU.
MATT STALLER GO AWAY!
HEYMRGELLER-MATH #2
REG YOU LOOK MAAAHVELOUS!
REG DON'T BE SO CONCEITED!
GREASED BOYS ARE
BODENHEIMER'S TOYS
THIS IS THE 17TH BOOSTER.
QEZ?ZEPHULUG!QED!SMAKSMAK
METS ARE NUMBER ONE!!!!!!
METS SUCK!!D-A-R-Y-L!!
KKIL AND ZIGGY-BOBBY LIU
YANKEES ROCK
Mets are nutriently @#$!
Slimer is the coolest!
Mets are the champs! #1
MATH TEAM IS FUN!!
YANKS VS METS = YANKEES NA
JETS ARE GOING ALL THEWAY
TONY AND JERRY-BLOOD BROS
Ματη Συρϖεψ ισ α γαψ μαγ.
Mrs Abramson is beautiful
Girls socker rocks!!!!!!!!!
BOSTON REDSUX

DIE COGNITIVE SCIENCE!!
DIE PAINFULLY LINGUISTS!
MATT ALEXANDER WILL BURN!
BURN I SAY! BURN FOREVER!
WHY, MIKA, WHY?
METS #1
STUYVESANT RULES!
MATH!
METS AND YANKEES IN '87
I LOVE THE NEW YORK METS!
HA HA HA YOU'RE ALL WEIRD!
ESTHER - HE'S LAUGHING!
GEN STRIKES A-GEN!!!!!!!!
YANKEES IN '87
!!!!HELLO!!!!
YANKEES IN '87? HA!
RAHUL IS A MACHO MAN
JON IS A PERVERT
DEAR SHELL, BEST FRIENDS
FOREVER-LOVE YA, JEANNIE
THIS LINE LEFT BLANK...
KCUF - There! I said it!
NOYTENGOPNADA1PUEDO7DICER
DERIVATIVE WE FALL.
INTEGRAL WE STAND.
THE BEST ONES ARE THE GOOD ONES.
Don't read this sentence.
Elizabeth Wilmer is SOO sexy!
GO MET'S

GREAT YANKEES OF 1986

the null set

NEW YORK UNIVERSITY
DEPARTMENT OF MATHEMATICS
PRESENTS:

... Two fascinating brain teasers to test your mettle. We invite you, and the Bronx High School of Science, to submit your solutions for a splendid award. We will evaluate all attempts, even if incomplete. Good Luck!

1. Suppose n cars drive into town and intend to park on the same block. The block contains exactly n parking spaces in a single file, so of course it will be easy for all the cars to park. However, each of the n drivers has a first choice of a parking spot without knowing the other driver's choices. If, when a driver arrives at his chosen spot, it is already occupied he moves on to the next unoccupied spot, if available, to park. What is the probability that all drivers park successfully? Note: All parking takes place on a one-way street, so the drivers may not back up to an unoccupied spot.

2. We play the following game. Initially n poker chips are placed at the 0 point on the number line. Next, if n is even, we divide the pile in half and place half the pile at -1 and the other half at $+1$. On the other hand, if n is odd, we leave one chip alone and split the remaining $n-1$ chips in half as before. We continue in the same way with the resulting piles. For example, if $n=6$, we have the following pattern of chips

6
303
11211
12021
20202
1020201
1102011
1110111

after which there is obviously no further change.

(a) show that the game terminates in a finite number of moves.

Let $f(n)$ denote the number of moves in the game. By part (a), $f(n)$ is finite. Show that

(b) $f(2n) = f(2n+1)$.

(c) $f(n)/n^2$ is bounded (i.e., $f(n)/n^2 \leq c$ for some number c).

Send your solutions to: Mathematics Contest c/o Lori Smith
Department of Mathematics
New York University
251 Mercer Street
New York, NY 10012

Deadline: 7/7/87